

Sentiment Analysis with an RNN in Keras

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1. Overview

Key point: An RNN only reads numbers, so text must become numbers first. Two ways: integer encoding (each word becomes its index in the vocabulary) and a learned embedding (each word becomes a short, dense vector that the network learns). On IMDB movie reviews, the same SimpleRNN reaches about 0.80 test accuracy with an embedding, against 0.50 with raw integers.

Sentiment analysis is the task of predicting whether a text is positive or negative. The data is a set of texts, each with a label: 1 for positive, 0 for negative. This Note builds a sentiment analysis model with the RNN of the [RNN forward propagation Note](#), in Keras, on real movie reviews.

The aim is the workflow, not the best accuracy: how text becomes numbers, how the numbers enter a **SimpleRNN**, and what changes when a learned embedding is added.

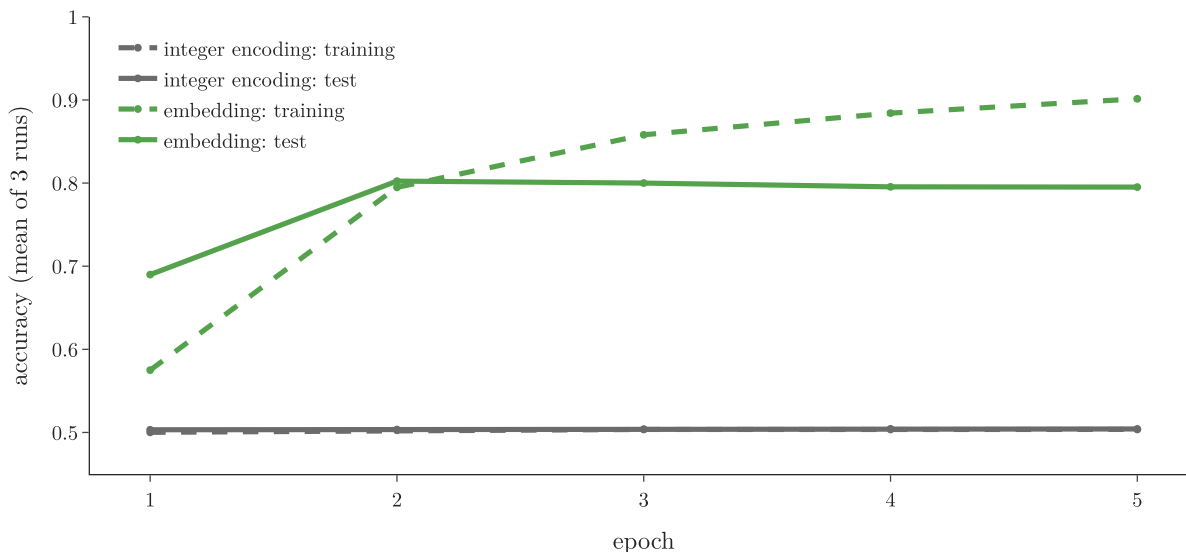


Figure 1: IMDB reviews cut to their last 50 words. Dashed: training accuracy; solid: test accuracy; mean of 3 runs. Raw integer inputs stay near a coin toss; a learned 2-number embedding lifts the same SimpleRNN to about 0.80 on the test reviews

Figure 1 is the result of the whole Note.

2. Prerequisites

- The [RNN forward propagation Note](#): input shape (time steps, input features), the recurrent layer, counting its parameters.
- The [why RNNs Note](#): padding sequences to a common length, and its cost.
- The [MNIST ANN Note](#): `compile`, `fit` and validation data in Keras.
- The [one-hot encoding Note](#): a category as a vector with a single 1.

3. Integer encoding

Key point: Build the vocabulary, give each word an integer, replace every word by its integer, then pad the sequences to one length.

3.1 The three steps

Key point: Vocabulary, replace, pad.

Each review is one **observation** (one record of the data), and its label is the **target** (the output we predict). Before an RNN can read the reviews, every word must become a number. **Integer encoding** does it in three steps.

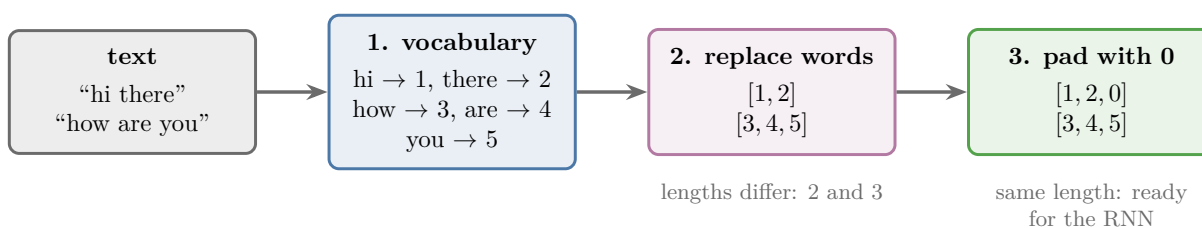


Figure 2: Integer encoding of two sentences: build the vocabulary and number its words, replace every word by its number, then pad the shorter sequence with 0

1. **Vocabulary.** List the unique words of the whole document, the **vocabulary**, and give each one an integer: hi is 1, there is 2, how is 3, are is 4, you is 5.
2. **Replace.** Write every sentence as the integers of its words: “hi there” becomes [1, 2] and “how are you” becomes [3, 4, 5].
3. **Pad.** The sequences have different lengths, so add zeros at the start or the end of the shorter ones until all have the same length: [1, 2, 0] and [3, 4, 5]. The integer 0 is kept for padding, so no word gets it.

Figure 2 shows the three steps.

3.2 Tokenizing in Keras

Key point: Keras’ `TextVectorization` layer lower-cases the text, strips punctuation, splits it into words, builds the vocabulary and replaces each word by its index, in one step.

Splitting a text into words is **tokenization**. Doing it by hand means lower-casing every word, removing punctuation and numbering the words; Keras does all of it.

Take a document of 10 short cheering slogans, such as “go india”, “india india” and “jeetega bhai jeetega india jeetega”. `adapt` reads the document and builds the vocabulary, most frequent word first. The Notebook prints 19 entries:

Index	Entry
0	"" (padding)
1	[UNK] (unknown word)
2	india
3	jeetega
...	15 more words

“india” appears 4 times, more than any other word, so it gets the first free index, 2.

Out-of-vocabulary words. A word seen only at prediction time, never in training, has no index. The layer maps every such word to a special **out-of-vocabulary (OOV) token**, [UNK], with index 1. In the Notebook, “go pakistan” becomes [16, 1]: “pakistan” is not in the vocabulary.

Python: Integer encoding with `TextVectorization`; it also pads the sequences with 0 at the end.

```
import keras
docs = ["go india", "india india", "hip hip hurray", ...]
vectorizer = keras.layers.TextVectorization()
vectorizer.adapt(docs)                # build the vocabulary
print(vectorizer.get_vocabulary())    # ['', '[UNK]', 'india', ...]
sequences = vectorizer(docs).numpy()  # one padded row per sentence
```

For lists of integers of different lengths, `keras.utils.pad_sequences(sequences, padding="post")` pads at the end, and `padding="pre"` (the default) at the start (Keras documentation).

Extra: Older Keras code uses `keras.preprocessing.text.Tokenizer` with `fit_on_texts`, `word_index`, `word_counts` and `texts_to_sequences`, and an `oov_token` argument. Keras 3 keeps that class only as a legacy API; `TextVectorization` is its replacement, and it reserves index 0 for padding and index 1 for [UNK] (Keras documentation, `TextVectorization`).

4. The IMDB dataset

Key point: 50,000 movie reviews, half for training and half for testing, already integer encoded. We pad or cut every review to 50 words.

4.1 Loading the reviews

Key point: `keras.datasets.imdb.load_data()` returns the reviews as lists of integers, one integer per word.

The **IMDB dataset** holds 50,000 movie reviews from the Internet Movie Database, labelled positive or negative: 25,000 for training and 25,000 for testing (Keras documentation, IMDB

dataset). Keras ships it already tokenized and integer encoded, so the steps of section 3 are done. The first training review starts

[1, 14, 22, 16, 43, 530, 973, 1622, 1385, 65, ...]

where 1 marks the start of a review and every other integer is a word. Decoded with the dataset's word index, it reads "this film was just brilliant casting location scenery story direction everyone's really suited the part they played and you ...".

Extra: In `load_data`, a word's integer is its frequency rank plus 3: 0 is padding, 1 is the start marker and 2 is the out-of-vocabulary word. So the most frequent word, "the", is 4 (Keras source, `imdb.load_data: start_char=1, oov_char=2, index_from=3`).

4.2 Cutting every review to 50 words

Key point: Reviews have different lengths (the first three have 218, 189 and 141 words). `pad_sequences(maxlen=50)` pads the short ones and cuts the long ones to 50 words, so training is fast.

Training on full reviews is slow, so we keep 50 words per review: `keras.utils.pad_sequences(X, maxlen=50)`. Shorter reviews get zeros in front; longer reviews are cut. The training data then has shape (25000, 50). The cut throws information away: the median review has 178 words. With time, use full reviews.

Which 50 words? `pad_sequences` cuts from the start by default (`truncating="pre"`), so a long review keeps its **last** 50 words, not its first 50 (Keras documentation, `pad_sequences`). Passing `truncating="post"` keeps the first 50.

5. Approach 1: integer encoding into a SimpleRNN

Key point: Feed each review as 50 time steps of 1 number each, into a SimpleRNN of 32 nodes and a sigmoid output node: 1,088 + 33 parameters.

5.1 The model

Key point: Input shape (50, 1): 50 time steps, 1 input feature, the word's integer.

Python: The model for raw integer inputs.

```
model = keras.Sequential([
    keras.Input(shape=(50, 1)),      # (time steps, input features)
    keras.layers.SimpleRNN(32, return_sequences=False),
    keras.layers.Dense(1, activation="sigmoid")])
```

Each review is 50 integers. At time step 1 the first integer enters the recurrent layer, at time step 2 the second, and so on, as in the [RNN forward propagation Note](#). After the 50th, the last hidden state goes to the sigmoid node, which gives $\hat{y}$.

5.2 Counting the parameters

Key point: $32 + 32 \times 32 + 32 = 1088$ in the recurrent layer and $32 + 1 = 33$ in the output layer.

Part	Count
W_i : 1 input feature to 32 nodes	$1 \times 32 = 32$
W_h : feedback, 32 nodes to 32 nodes	$32 \times 32 = 1024$
b_h	32
SimpleRNN total	1,088
W_o and b_o	$32 + 1 = 33$

`model.summary()` prints the same numbers.

5.3 return_sequences

Key point: `return_sequences=False` (the default) returns only the last hidden state. `return_sequences=True` returns the hidden state of every time step.

The recurrent layer computes a hidden state at every time step. With `return_sequences=False`, only the last one leaves the layer; the others stay inside and only feed the next time step. Sentiment analysis needs one answer per review, after the last word, so `False` is right here.

Some tasks need an output at every word: named entity recognition labels each word (for example, is it a person's name?), and machine translation produces a sentence. Those use `return_sequences=True`. The [types of RNN Note](#) covers these cases.

5.4 Training

Key point: Binary cross-entropy, Adam, 5 epochs, the test reviews as validation data. Raw integers give a test accuracy near a coin toss.

We compile with binary cross-entropy loss and the Adam optimizer, train for 5 epochs and pass the test reviews as `validation_data`. Averaged over 3 runs, the model reaches 0.50 training accuracy and 0.50 test accuracy after 5 epochs (grey lines in Figure 1). With two balanced classes, guessing gives 0.50.

Only 50 words per review and only 5 epochs make the task harder. But the embedding model of section 7 gets the same 50 words and the same 5 epochs and reaches 0.80. The main difference between the two models is how a word enters the RNN (the embedding model also keeps only the 10,000 most frequent words), so the representation of the words is what holds this model back.

6. Word embeddings

Key point: An embedding turns each word into a short vector of real numbers, most of them non-zero, learned so that words used in similar ways get similar vectors.

6.1 Sparse and dense representations

Key point: Padding and one-hot vectors are mostly zeros (sparse). An embedding is short and full of useful numbers (dense).

Two problems come with the earlier representations.

- **Sparse.** Take a 20-word review padded to the longest review of 2,000 words: 1,980 of its 2,000 numbers are padding zeros. A one-hot vector over a 10,000-word vocabulary is 9,999 zeros and one 1. A representation where most values are 0 is **sparse**.
- **No meaning.** Integers and one-hot vectors say nothing about meaning. In one-hot space every pair of words is the same distance apart, $\sqrt{2}$, so “good” is as far from “great” as from “awful” (Goodfellow §12.4.2).

A **word embedding** represents each word as a real-valued vector of a chosen, small size, such as 2 or 32 numbers, most of them non-zero: a **dense** representation. The vectors are learned so that words that appear in similar contexts are close to each other, which often puts words with similar meanings next to each other (Goodfellow §12.4.2).

6.2 Learning the embedding with the model

Key point: Keras’ **Embedding** layer learns the vectors during training, by backpropagation, together with the RNN.

Word2Vec and GloVe are well-known embedding methods trained on large text collections. In deep learning we can also learn an embedding as part of our own model: Keras’ **Embedding** layer turns each integer into a dense vector of fixed size (Keras documentation, **Embedding**). Its input must be integer encoded.

Embedding takes two main arguments:

- `input_dim`: the vocabulary size;
- `output_dim`: the size of each word’s vector, a hyperparameter (2, 32, 200, ...) to tune.

Extra: Older code also passes `input_length`, the number of words per sequence. Keras 3 removed that argument; the sequence length comes from the input shape (`keras.Input(shape=(50,))`) instead (Keras documentation, **Embedding**).

6.3 What the layer computes

Key point: The layer holds a weight matrix E with one row per word. A word's vector is its row of E : a one-hot vector times E , done as a lookup.

1. **In words:** the layer is like a dense layer whose input is the word's one-hot vector, with no bias and no activation. Multiplying a one-hot vector by a matrix picks one row, so the layer simply looks up row k for word k .
2. **Formula:** for a vocabulary of V words and vectors of size d , E is $V \times d$, and

$$\text{embedding}(k) = \text{onehot}(k) E = E_{k,:}$$

3. **Example:** the slogan document has $V = 19$ entries. With $d = 2$, E is 19×2 : 38 weights, which `model.summary()` confirms. The first word of the first slogan, “go”, has index 16, and row 16 of E is its 2-number vector: $[-0.039, -0.006]$ before training.

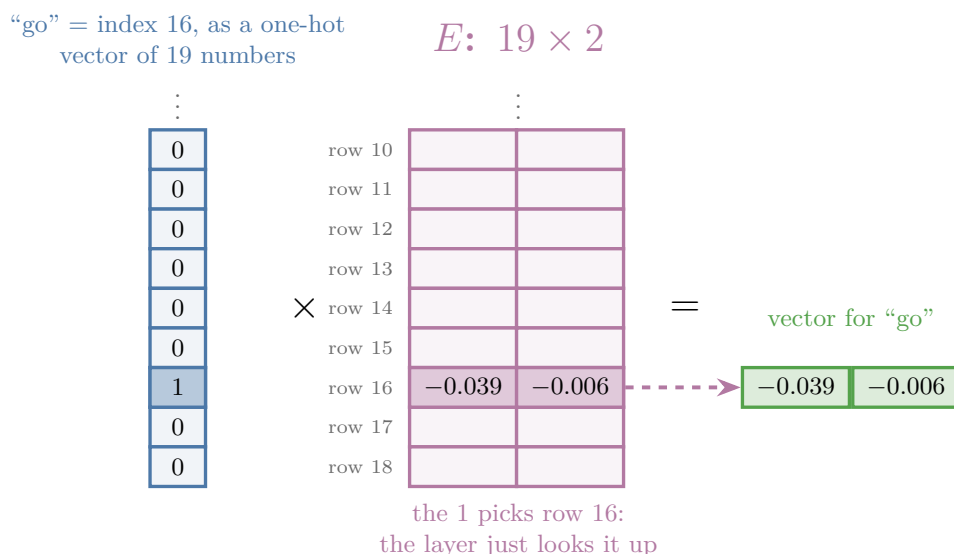


Figure 3: The embedding layer as a lookup: the one-hot vector of “go” (index 16) times E keeps row 16 of E , which becomes the word’s dense vector

E starts random and is trained with the rest of the network by backpropagation. Figure 3 shows the lookup. The Notebook checks that the one-hot product and the layer’s output are the same numbers.

Applied to a padded sequence of 5 integers, the layer returns 5 rows of 2 numbers, one per word (the padding index 0 also gets a row). For the 10 slogans, padded to 5 words, the output has shape $(10, 5, 2)$: 100 numbers.

7. Approach 2: an embedding into the same SimpleRNN

Key point: Keep the 10,000 most frequent words, embed each in 2 numbers, and feed the 50 vectors into the SimpleRNN. Test accuracy rises to about 0.80.

7.1 The model

Key point: Embedding (20,000 parameters), SimpleRNN (1,120), Dense (33).

`load_data(num_words=10000)` keeps the 10,000 most frequent words; every rarer word becomes the out-of-vocabulary integer 2.

Python: The embedding model.

```
model = keras.Sequential([
    keras.Input(shape=(50,)),          # 50 integers per review
    keras.layers.Embedding(10000, 2),  # each word -> 2 numbers
    keras.layers.SimpleRNN(32),
    keras.layers.Dense(1, activation="sigmoid")])
```

Now each time step carries 2 numbers instead of 1, so the RNN's input is (50 time steps, 2 input features).

Layer	Parameters
Embedding: E	$10000 \times 2 = 20000$
SimpleRNN: W_i, W_h, b_h	$2 \times 32 + 32 \times 32 + 32 = 1120$
Dense: W_o, b_o	$32 + 1 = 33$

7.2 Results

Key point: Same RNN, same 50 words, same 5 epochs: the embedding lifts test accuracy from 0.50 to 0.80. Training accuracy climbs higher still, a sign of overfitting.

After 5 epochs, mean of 3 runs	Integer encoding	Embedding
Training accuracy	0.50	0.90
Test accuracy	0.50	0.80

The green lines of Figure 1 show the embedding model. Test accuracy peaks at epoch 2 (0.80) while training accuracy keeps rising to 0.90: the model is starting to memorise the training reviews, which is **overfitting**. The [regularization Note](#) and the [early stopping Note](#) cover the remedies.

7.3 What the embedding learned

Key point: After training, positive and negative words sit in different parts of the 2-number space.

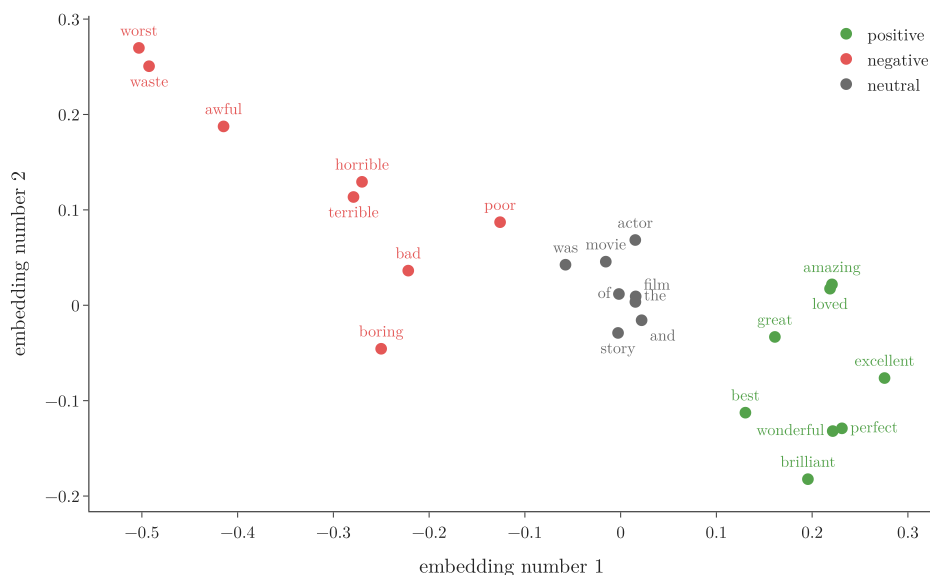


Figure 4: The learned 2-number vectors of a few IMDB words after training. Positive words (green) and negative words (red) end up on opposite sides; frequent neutral words (grey) sit between them

Figure 4 plots the learned vectors of 8 positive, 8 negative and 8 neutral words. Nobody told the network which words are positive. Goodfellow §12.4.2 explains the pattern: words that share features learned by the model end up close together. The only label this model learns from is the sentiment, and Figure 4 shows its embedding sorting the words by sentiment along one direction.

7.4 Learned or pre-trained embeddings

Key point: We can learn the embedding with the model, as here, or load pre-trained vectors such as Word2Vec or GloVe.

A pre-trained embedding was learned on a large general text collection. An embedding learned with the model is fitted to our own data and task. Which works better depends on the dataset.

8. Summary

	Integer encoding	Embedding
A word becomes	1 number, its index	a learned vector of d numbers
Representation	arbitrary numbers, sparse after padding	dense, similar words close
RNN input shape	(50, 1)	(50, d)
Extra parameters	none	vocabulary size $\times d$
IMDB test accuracy, 5 epochs	0.50	0.80

- Text must become numbers: build a vocabulary, replace words by integers, pad to one length.
- `TextVectorization` tokenizes and integer encodes; unknown words map to [UNK]; `pad_sequences` pads and cuts.
- IMDB comes integer encoded; `pad_sequences(maxlen=50)` keeps the last 50 words of each review.
- `SimpleRNN(32)` on 1 input feature has 1,088 parameters; `return_sequences=False` returns only the last hidden state.
- An **Embedding** layer is a lookup table E , trained with the model; it turned a coin toss into about 0.80 accuracy.

9. Sources

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. §12.4.2 (neural language models, word embeddings). deeplearning-book.org/contents/applications.html.
- Keras API documentation: `TextVectorization`, `pad_sequences`, `Embedding`, `SimpleRNN` layers, and the IMDB dataset (`keras.datasets.imdb.load_data`), keras.io/api/. Checked against the installed Keras 3.15 source.
- Maas, A. L. et al. (2011). Learning Word Vectors for Sentiment Analysis. ACL 2011. (The IMDB dataset.)

10. Key terms

Term	Meaning
Sentiment analysis	Predicting whether a text is positive or negative
Observation	One record of the data, here one review
Target	The output we predict, here the sentiment label
Vocabulary	The set of unique words in the data, each with an integer index

Term	Meaning
Tokenization	Splitting a text into words (tokens)
Integer encoding	Replacing each word by its index in the vocabulary
Out-of-vocabulary (OOV) token	A placeholder, [UNK], for words not in the vocabulary
Padding	Adding zeros to sequences so that all have the same length
Sparse representation	A representation where most values are 0
Dense representation	A short representation where most values are non-zero
Word embedding	A learned real-valued vector for each word; words used in similar ways get nearby vectors
Embedding layer	The Keras layer holding the embedding matrix E ; looks up one row per word
<code>return_sequences</code>	SimpleRNN argument: False returns the last hidden state, True returns every hidden state
IMDB dataset	50,000 labelled movie reviews, shipped with Keras already integer encoded
Overfitting	Training accuracy rising while test accuracy stalls or falls